

Univariate Q&A:

1.What are the summary statistics of the dataset?

```
#dataset  
df.describe()
```

Note:

- 1.Displays mean, median, minimum, maximum and quartiles
- 2.Helps to understand the overall data distribution

2.Are there any missing values?

```
df.isnull().sum()
```

Note:

- 1.Shows missing values in each column
- 2.After preprocessing, all values should be 0

3. What are the data types of the variables?

```
df.info()
```

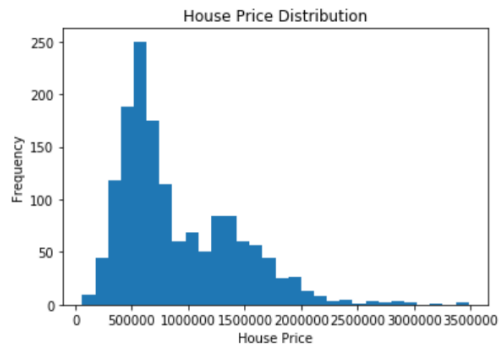
Note:

- Displays data types and non-null counts
- Ensures numerical and categorical columns that are identified correctly

4. What is the distribution of house prices?

```
import matplotlib.pyplot as plt

plt.hist(df['house_price'], bins=30)
plt.title('House Price Distribution')
plt.xlabel('House Price')
plt.ylabel('Frequency')
plt.show()
```



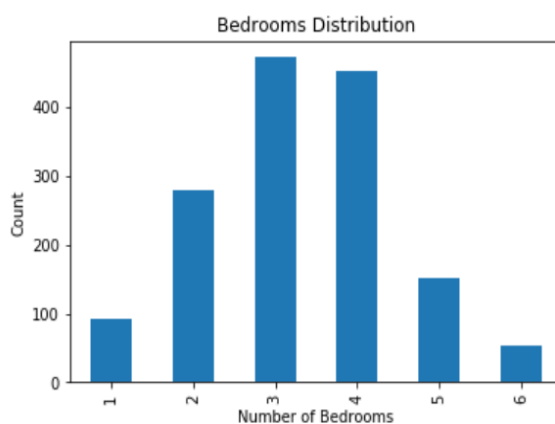
Note:

- The histogram shows how house prices are distributed
- Most houses fall within a specific range
- To check whether the distribution is symmetric or skewed

5. How are the bedrooms distributed?

```
import matplotlib.pyplot as plt

df['bedrooms'].value_counts().sort_index().plot(kind='bar')
plt.title('Bedrooms Distribution')
plt.xlabel('Number of Bedrooms')
plt.ylabel('Count')
plt.show()
```



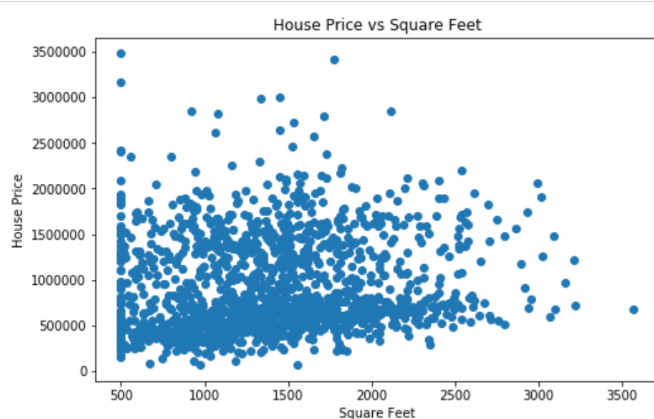
Note:

- The bar chart shows the number of houses with each bedroom count
- The tallest bar tells the most common number of bedrooms

Bivariate Q&A:

1. Is there a relationship between square feet and house price?

```
#Bivariate analysis
plt.figure(figsize=(8,5))
plt.scatter(df['square_feet'], df['house_price'])
plt.title('House Price vs Square Feet')
plt.xlabel('Square Feet')
plt.ylabel('House Price')
plt.show()
```

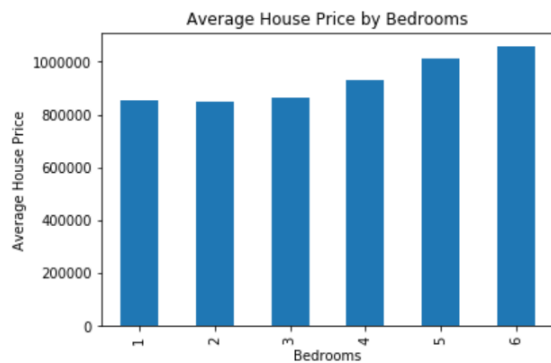


Note:

- Houses with larger square footages generally have higher prices.
- A Positive relationship can be observed.

2. What is the average house price of each bedroom category?

```
df.groupby('bedrooms')['house_price'].mean().plot(kind='bar')
plt.title('Average House Price by Bedrooms')
plt.xlabel('Bedrooms')
plt.ylabel('Average House Price')
plt.show()
```

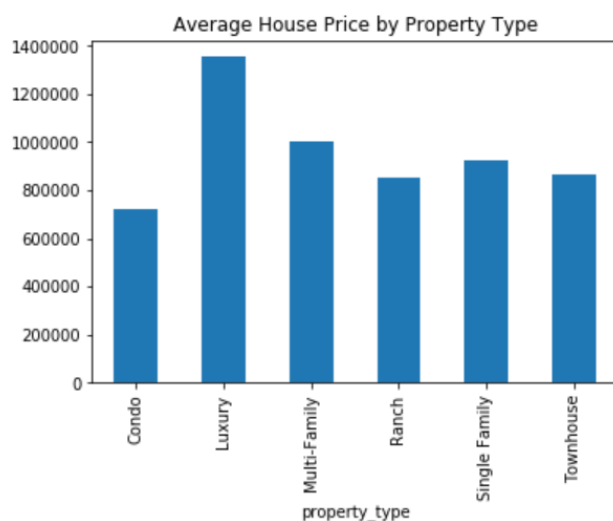


Note:

- Compares the average price across the different bedroom counts
- More bedrooms lead to higher average prices

3. What is the average house price by property type?

```
df.groupby('property_type')['house_price'].mean().plot(kind='bar')
plt.title('Average House Price by Property Type')
plt.show()
```



Note:

- Compare average prices across property types
- Identifies which property type has the highest average price

4. Which features are most correlated with house price?

```
df.select_dtypes(include=['number']).corr()['house_price'].sort_values(ascending=False)
```

```
house_price      1.000000
square_feet      0.122299
bathrooms        0.120774
bedrooms         0.104466
lot_size_sqft    0.096001
year_built       0.032189
nearby_school_rating 0.022183
days_on_market  0.015660
garage_spaces    -0.011799
crime_rate_index -0.018784
parking_spaces   -0.026760
distance_to_city_center_km -0.124433
Name: house_price, dtype: float64
```

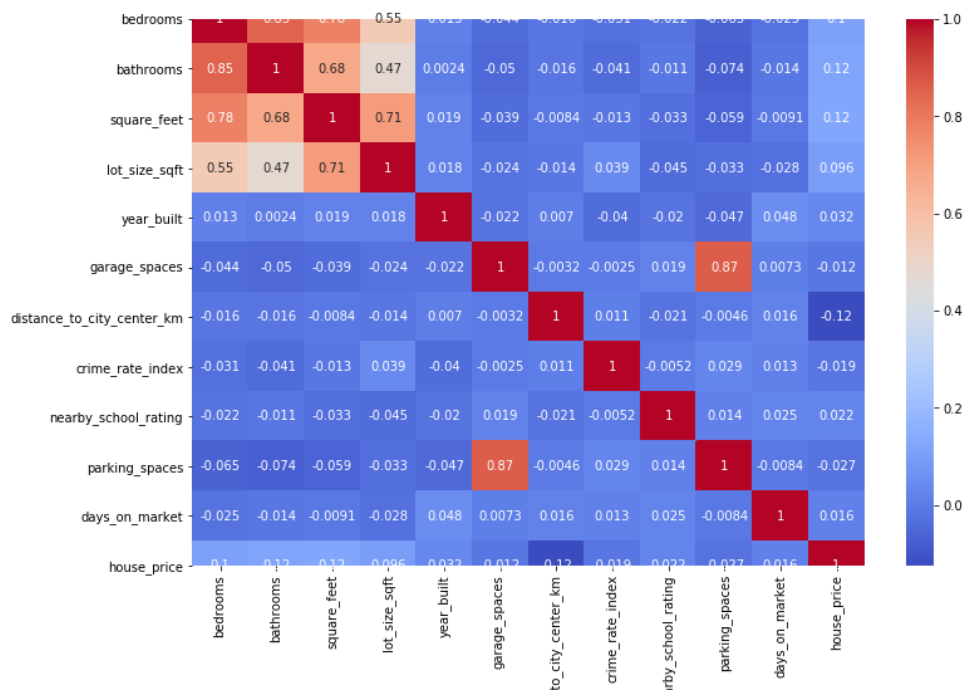
Note:

- Shows the correlation of each numerical feature with house price
- Features with high positive correlation are more influential

5. What is the correlation among numerical features?

```
: import seaborn as sns
import matplotlib.pyplot as plt

plt.figure(figsize=(12,8))
sns.heatmap(
    df.select_dtypes(include=['number']).corr(),
    annot=True,
    cmap='coolwarm'
)
plt.show()
```



Note:

- The heatmap visualizes relationships among all numerical variables.
- Stronger positive relations in dark positive values and negative values remind inverse relationship